

waph-munozsa

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Student

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Short-bio: Sophia Munoz has interests in Computer Graphics and Web Application development.



Figure 1: Sophia's headshot

Lab 0 - Development Environment Setup

The Lab's Overview

For Lab0 there were two parts. Part I is setting up the Ubuntu 22.04 Virtual Machine through VirtualBox and installing all the applications necessary (e.g. net-tools, apache web server, git, sublime, pandoc, pdflatex, and Google Chrome). Part II entailed cloning the course repository and creating a private repository. This included following the git instructions to successfully complete the Lab0 report.

Lab0 Folder: <https://github.com/munozsophia/waph-munozsa/blob/main/labs/lab0>.

Part I - Ubuntu Virtual Machine & Software Installation

Ubuntu 22.04 Virtual Machine Setup Follow the instructions on Ubuntu VM Tutorial

- Install and configure VirtualBox
- Import an Ubuntu image
- Run a virtual instance of Ubuntu Desktop
- Setup NAT and Host-only Adapters

net-tools Intallation Now that the VM has an IP address accessible within the virtual network

At the command prompt, type `sudo apt install net-tools`, type `ifconfig` to view IP address

Apache Web Server Intallation At the command prompt, type `$ sudo apt install apache2`

git Installation At the command prompt, type `$ sudo apt install git`

Sublime Text Editor Installation At the command prompt, type `$ sudo snap install sublime-text --classic`

pandoc Installation At the command prompt, type `$ sudo apt install pandoc`

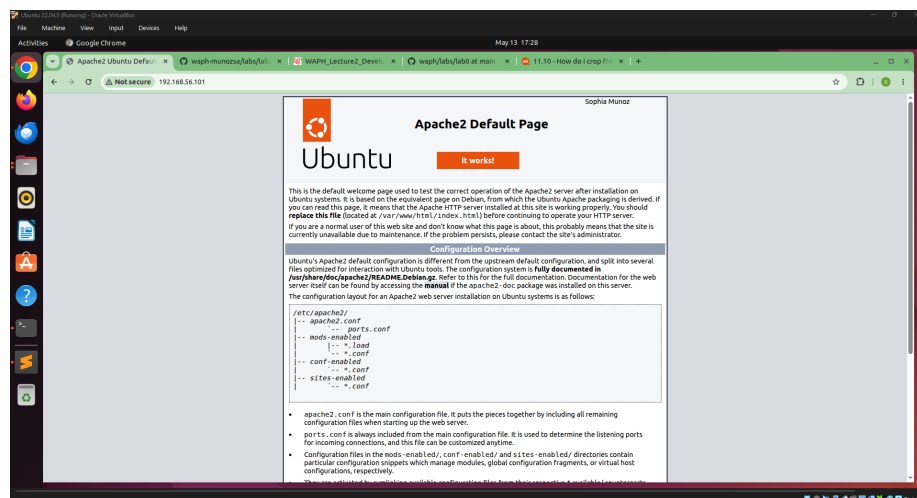
pdflatex Installation Since `pandoc` requires `pdflatex` and its fonts to render files, type in the following:

- `$ sudo apt-get install texlive-latex-base`
- `$ sudo apt-get install texlive-fonts-recommended`
- `$ sudo apt-get install texlive-latex-extra`
- `$ sudo apt-get install texlive-fonts-extra`

Google Chrome Installation

- Download Google Chrome (.deb package)
 - Open Firefox
 - Visit the Google Chrome download page
 - Select **64-bit.deb** (For Debian/Ubuntu)
 - Click **Install** button
- Open the Downloaded File
 - Go to **Downloads**
 - Double-click `google-chrome-stable_current_amd64.deb`
 - It will open in **Ubuntu Software Center**
- Install Google Chrome
 - Click Install
 - Enter your system password when prompted

Apache Web Server Testing



Apache Web Server in Ubuntu VM



Apache Web Server in Laptop

Part II - git Repositories and Exercises

Course Repository

```

user@Ubuntu22: ~/waph
user@Ubuntu22:~$ pwd
/home/user
user@Ubuntu22:~$ git clone https://github.com/waph-phung/waph.git
Cloning into 'waph'...
remote: Enumerating objects: 14, done.
remote: Counting objects: 100% (14/14), done.
remote: Compressing objects: 100% (12/12), done.
remote: Total 14 (delta 3), reused 6 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (14/14), 8.16 KiB | 928.00 KiB/s, done.
Resolving deltas: 100% (3/3), done.
user@Ubuntu22:~$ cd waph && git pull
Already up to date.
user@Ubuntu22:~/waph$

```

Figure 2: Cloned Course Repository

Cloned Course Repository

Private Repository

The steps I took to create my private repository on GitHub.com:

- Click the green **Create repository** button
- Name the repository with the following naming convention **waph-your-uc-username** (**waph-munozsa**)
- Check off Private configuration
- Select **Add a README file**
- Choose license **Apache License 2.0**

To share the repository with Instructor **phung-waph**

From the private repository, click on:

- settings -> collaborators -> add people
- type user **phung-waph** and add it

This repository's full URL is <https://github.com/munozsophia/waph-munozsa.git>.

Generating and Setting up SSH Keys

- From the terminal: **\$ ssh-keygen**
- Press Enter: accepting default location
- Enter/re-enter a passphrase when terminal asks (can be blank too)
- The key files are stored in **/.ssh** folder. To view the files: **\$ ls ~/.ssh**
- To view and copy the key: **\$ cat ~/.ssh/id_rsa.pub**
- Then select and copy

To add the public key to a GitHub account:

- Click account profile
- Click settings -> SSH and GPG keys -> new ssh key
- Title the key and paste the copied key to the Key section
- click **Add SSH key**

Clone Private Repo to VM using SSH Key In home directory (~), **\$ git clone <url>** (**git@github.com:munozsophia/waph-munozsa.git**)

Get URL from the repo overview, click Code -> SSH

Committed changes to Private Repository

```
user@Ubuntu22: ~/waph-munozsa
user@Ubuntu22:~/waph-munozsa$ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 383 bytes | 383.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:munozsophia/waph-munozsa.git
   788f40e..df7e73a  main -> main
user@Ubuntu22:~/waph-munozsa$ git add .
user@Ubuntu22:~/waph-munozsa$ git commit -m "Updated README.md headshot file"
[main 9d4f46a] Updated README.md headshot file
 2 files changed, 1 insertion(+), 1 deletion(-)
   rewrite images/headshot.jpg (99%)
user@Ubuntu22:~/waph-munozsa$ git pull
Already up to date.
user@Ubuntu22:~/waph-munozsa$ git push
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 4 threads
Compressing objects: 100% (4/4), done.
Writing objects: 100% (5/5), 13.20 KiB | 13.20 MiB/s, done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To github.com:munozsophia/waph-munozsa.git
   df7e73a..9d4f46a  main -> main
user@Ubuntu22:~/waph-munozsa$
```

Figure 3: Committed changes to Private Repository